William Thomas Campagna

Profile

I am currently completing my B.S. in Physics and working part time as a peer tutor. I work closely with Dr. Eugenii Donev performing research in photonics and lead an independent research project applying model predictive control and physics-informed machine learning to photovoltaic panel orientation.

Education

Austin Peay State University (APSU), BS in Science August 2023 – Present

- Major: Physics
- Minor: Mathematics
- Degree Honors GPA: 3.046
- Overall GPA: 2.956
- **Relevant Coursework:** AC/DC Circuits, Differential Equations, Linear Algebra, Modern Physics, Optics, Experimental Methods, Numerical Analysis, Statistical Methods, Quantum Mechanics

Work Experience

Peer Tutor, Center for Academic Support and Enrichment – Clarksville, TN Oct. 2025 - Present

- Provide tutoring three times a week for Precalculus Algebra, Precalculus, and Calculus 1.

Event Leader/Guide, Govs Outdoors – Clarksville, TN May 2024 – Oct. 2026

- Managed Govs Outdoors programs for rock climbing, hiking, camping, and other adventure-based coordinated trips.
- Presented *From the Classroom to the Real World: Rock Climbing* at the TIRSA Spring 2025 Conference hosted by **APSU**.

Mentor, Governors School for Computational Physics July 2026 - August 2026

- Over the course of a month I ran office hours, graded assignments and otherwise ensured the safety of the students of the Governors School for Computational Physics.

Camp Counselor, Charleston County Parks and Recreation – Mt. Pleasant, SC May 2023 - August 2023

- Performed regular duties, including managing communications between other counselors.
- Organizing educational activities and providing a fun summer experience for children ages 6 to 9 years.

Volunteer Experience

Physics, Engineering, Astronomy Outreach Events, Del Square Psi

- The club runs booths for schools and science events in which we demonstrate the magic of physics along with brief explanations on the fundamental forces at play.
- Demos Used: Tesla coil experiments, quantum locking demonstrations, angular momentum with a bicycle wheel.
- The club helps run public observation nights at the APSU observatory in which we set up telescopes, locate celestial bodies, and provide information about what we are observing.

Dream Team – Lancaster, SC

- Gave young children with learning disabilities the opportunity to play baseball with college athletes.

Community Service in the Dominican Republic 2021

- Performed a 3 km long beach sweep to help restore the environment. This included building and painting a house, providing clean water and food to people in need, and spending time getting to know the locals through baseball.

Leadership Experience

President of Emerald Hill Hall Council 2024-2025

- I hosted biweekly meetings with my fellow council members. In these meetings we coordinated monthly events for the school while planning around our yearly budget.

Public Outreach Officer for Del Square Psi 2024-2025

- As a Public Outreach Officer for Del Square Psi, my role includes communication and planning for future events. I try to be ambitious in my role so that the STEM field can be accessible to everyone.

Founder of the Del Square Psi Video Production Team 2025-Present

- As the founder and current leader of the DSP Video Production Team I manage a group of students in research and production of educational videos for various physics demos and concepts.

Guided Hiking and Rock Climbing Trips 2024-2025

- I stay detail oriented when preparing to lead a guided trip. This includes detailed research about the area, holding meetings and safety briefs, providing demonstrations as well as presentations on the environment.

Research Experience

Comparative Analysis on the Refractive Index of a Prism Using a Versatile Spectrometer-Goniometer Setup 2025-Present

- J. Martinez, K. Scott, **W. T. Campagna**, M. Graff, and E. U. Donev. In preparation for *Eur. J. Phys.*
- Compares seven geometric, intensity-based, and phase-sensitive methods across five laser wavelengths on a single spectrometer-goniometer setup; the angle of minimum deviation agrees with Sellmeier values to a maximum error of 0.048%.
- Co-designed the experimental schematics for the paper.

Model Predictive Control: Physics-Informed Surrogate for Photovoltaic Panel Orientation 2025-Present

- Independent research funded by the APSU Department of Physics, Engineering, and Astronomy, developing a model predictive controller for a 3-axis PV mount and a Raspberry Pi Pico data acquisition system.
- Trained a neural surrogate of a lumped thermal model on 10^6 rows from a Fortran RK4 generator; the surrogate matches its generator to 0.06 K and retrains in about 2 minutes on a laptop CPU.
- Showed that under a power objective the optimal pose is always the max-irradiance pose, making the 15-minute panel temperature forecast the surrogate's key output.
- Validated against a year of Sandia National Laboratories module data: a physics-calibrated model beats persistence (2.21 K vs. 3.20 K RMSE); in a closed-loop sandbox the controller captures 98.7% of oracle energy.

The Effects of Exercise on the Central Nervous System 2021-2022

- Research was funded by a program through the East Cooper Center for Advanced Studies.
- The research included a systematic review of six publications regarding the topic.

Presentations

Model Predictive Control: Physics-Informed Surrogate for Photovoltaic Panel Orientation November 2026

- Poster presented at the Southeastern Section of the American Physical Society (SESAPS) meeting, hosted by Kennesaw State University.

A Preliminary Study in the Effectiveness of Model Predictive Control in Single Junction Photovoltaic Panels April 11th, 2026

- An introduction to this research at The Middle Collegiate Division Meeting of the Tennessee Academy of Science.

Comparing Six Methods for Determining the Refractive Index of a Glass Prism Using a Multi-Wavelength Laser and a Versatile Optical Setup March 27-28th, 2026

- An update to our research at the Tennessee American Association of Physics Teachers, hosted by the University of Tennessee Chattanooga.

Neutrino Oscillations: What Does it Mean December 9th, 2025

- A literature review on neutrino oscillations and the history of neutrinos alongside Eden Kope at the Austin Peay

State University Physics, Engineering, and Astronomy Symposium.

Comparing Five Methods for Determining the Refractive Index of a Glass Prism Using a Multi-Wavelength Laser and a Versatile Optical Setup October 31st, 2025

- An update to the research alongside Keira Scott in Denver, Colorado at the Society of Physics Students Conference.

The Effects of Exercise on the Central Nervous System May 8th, 2022

- A literature review on the neurological impact of weightlifting and running at the East Cooper Center for Advanced Studies.

Additional Experience

Designing and developing a database to handle the Austin Peay State University Astrophysics Lab's growing requirements 2026-Present

- **Code Languages Used:** C, Python, SQL

Other Projects

Game Developer, Train Wreck Company 2026-Present

- Program gameplay and engine systems in C++ for *BREACHPUNK*, a 2D narrative game in development.
- Develop the studio's in-house engine and editor used to prototype mechanics, playtest, and build its games.
- trainwreckcompany.github.io

Awards and Certifications

- **1st place in the 2023 Great Green Ideas Contest at APSU** 2023
- **Project Lead The Way, White Lab Coat Award** May 2022

Organizations

Society of Physics Students Sept. 2025-Present

- As a member of the Society of Physics Students I have attended the 2025 SPS Conference and worked with a selection of students to produce a video documentary highlighting the research and events at the conference.

Sustainable Campus Fee Committee Member Feb. 2024- Aug. 2025

- Being a member of the Campus Fee Committee put me in a position to vote on how the budget is spent towards making a greener and more inclusive campus.

Relevant Skills

Soft Skills:

- Strong ability to communicate complex ideas
- Comfortable working in a team environment
- Creative and adaptable use of problem solving skills

Software:

- Proficiency in the Microsoft Environment (Word, Excel, Powerpoint, etc.)
- Matlab (Intermediate)
- KiCAD (Intermediate)
- PyTorch (Intermediate)

Programming Languages:

- Python (Advanced)
- Fortran (Intermediate)
- SQL (Intermediate)
- C (Novice)
- C++ (Intermediate)